



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MECHATRONIC SYSTEMS
NQF LEVEL 2

3 March 2025

This marking guideline consists of 7 pages.

tvempapers.co.za **ultradeep.co.za**



Search for NCV Exam Papers



TVET Exam Papers

tvetpapers.co.za
UltraDeep.co.za
MyNATED.co.za

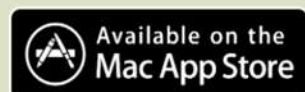
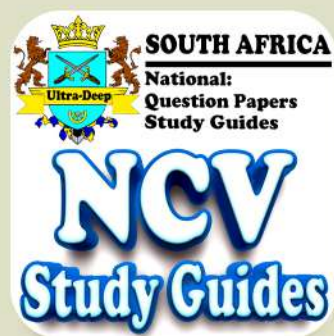
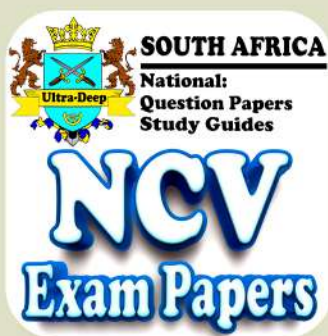


THANK YOU FOR DOWNLOADING THIS PAST EXAM PAPER/MEMO, WE HOPE IT WILL BE OF HELP TO YOU

These are the only apps affiliated with tvetpapers.co.za

Download Papers from
tvetpapers.co.za or MyNATED.co.za

These are some of our apps (visit tvetpapers.co.za)



Start your TVET journey with the best apps to help you achieve the best results in your exams. Our apps and websites cater for Nated and NCV students



Click on the banner above to visit tvetpapers.co.za or UltraDeep.co.za

DISCLAIMER

tvetpapers.co.za, mynated.co.za and ultradeep.co.za are part of Ultra-Deep Multimedia™ as such, no one; business or individual; are permitted to use (publish, display, transmit or broadcast) them in any other app/website/server/medium.

Copyright © 2024 - 2025 Ultra-Deep Multimedia - All Rights Reserved

QUESTION 1

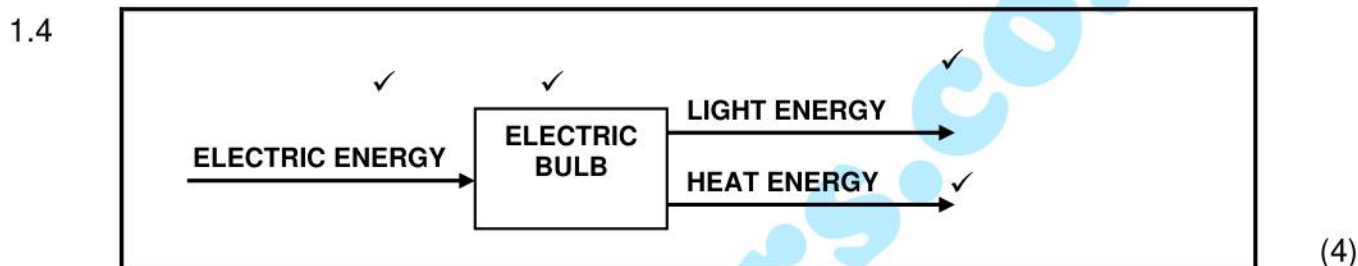
1.1 Energy conversion is energy changing from one form to another form. (2)

1.2 To energise or de-energise the system

OR

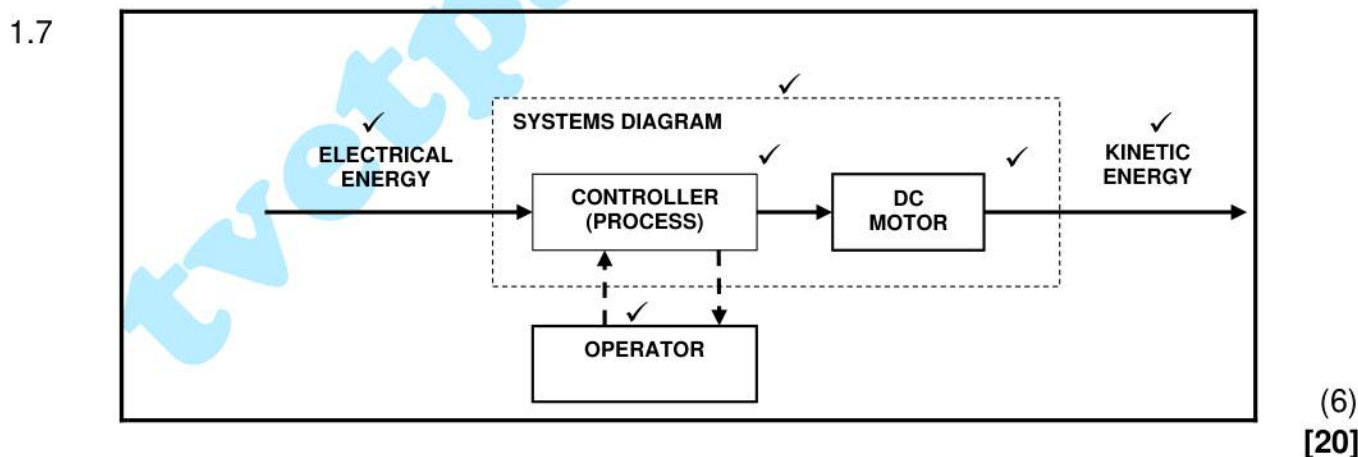
To turn the system on or off (2)

1.3 When the indication lamp is switched on, it indicates that the system is energised and when it is off, it indicates that the system is de-energised. (2)



1.5 Electric energy at the input is converted to magnetic lines of force (magnetic energy) which, in turn, is converted to kinetic energy at the output. (2)

- 1.6
- Understanding the operation of the system
 - Understanding the subsystems that make up the system
 - Assisting in troubleshooting the system
- (2)



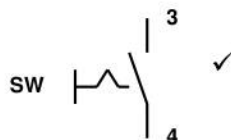
[20]

QUESTION 2

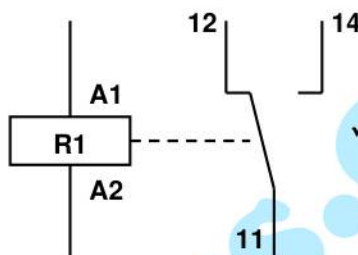
2.1 Indirect control

(1)

2.2 2.2.1

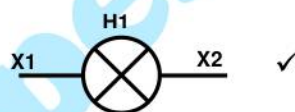
**Marks to be awarded if fully labelled**

2.2.2



Marks to awarded if fully labelled

2.2.3



Marks to be awarded if fully labelled

2.2.4

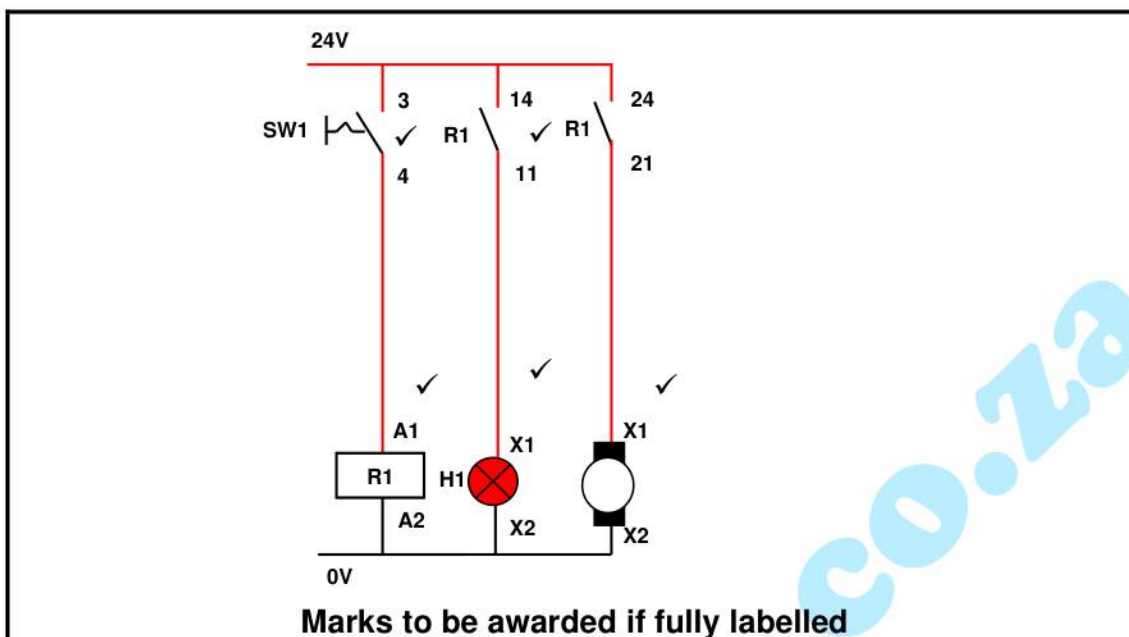


Marks to be awarded if fully labelled

(4 × 1)

(4)

2.3



(5)

2.4

- Hardwired control (relay ladder logic)
- Electronic control
- Embedded control
- Processor control (PLC/PC)

(4)

2.5

2.5.1 The solenoid actuator is a coil which, when energised, becomes a magnet and attracts a moving part known as the armature. (2)

2.5.2 The armature part of a relay is attached to contacts which are used to control electrical signals.
The armature part of a solenoid valve is connected to a spool which controls the pressurised air. (2 × 2)

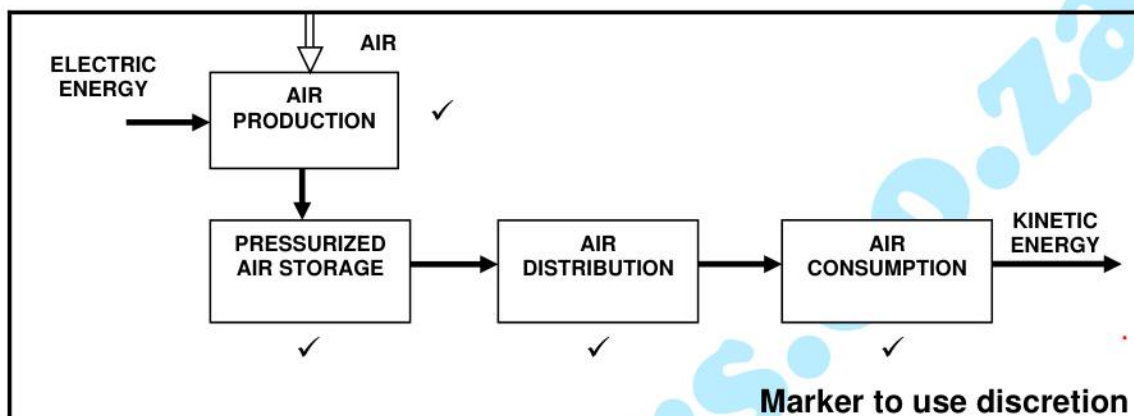
(4)
[20]

QUESTION 3

- 3.1
- Forming
 - Moving
 - Clamping
 - Sorting
 - Packaging

(Any 4 × 1) (4)

3.2



(4)

- 3.3
- Dead-end feed
 - Ring feed

(2)

- 3.4
- Insert the pipe into the connector by pushing the pipe into the connector.
 - To disconnect a pipe push the collar as shown and pull out the pipe.

(2 × 2) (4)

3.5 Basic rules that should be followed:

- The flow of working energy in a circuit diagram is drawn from the bottom to the top of the diagram.
- The sequence of the working cycle is from left to right of the circuit diagram; the air supply is therefore shown at the lower left, with the cylinder that performs the first stroke of the cycle being situated in the upper left.
- Power valves are drawn directly below the cylinders they operate.
- All circuit components are shown in their rest position, with the supply under pressure.
- Electrical connections are not generally shown.
- Mechanically operated valves, controlling the rest position of cylinder-driven parts, are operated in the rest position and must be drawn accordingly; external connections are drawn to the valve symbol square on the operator side.

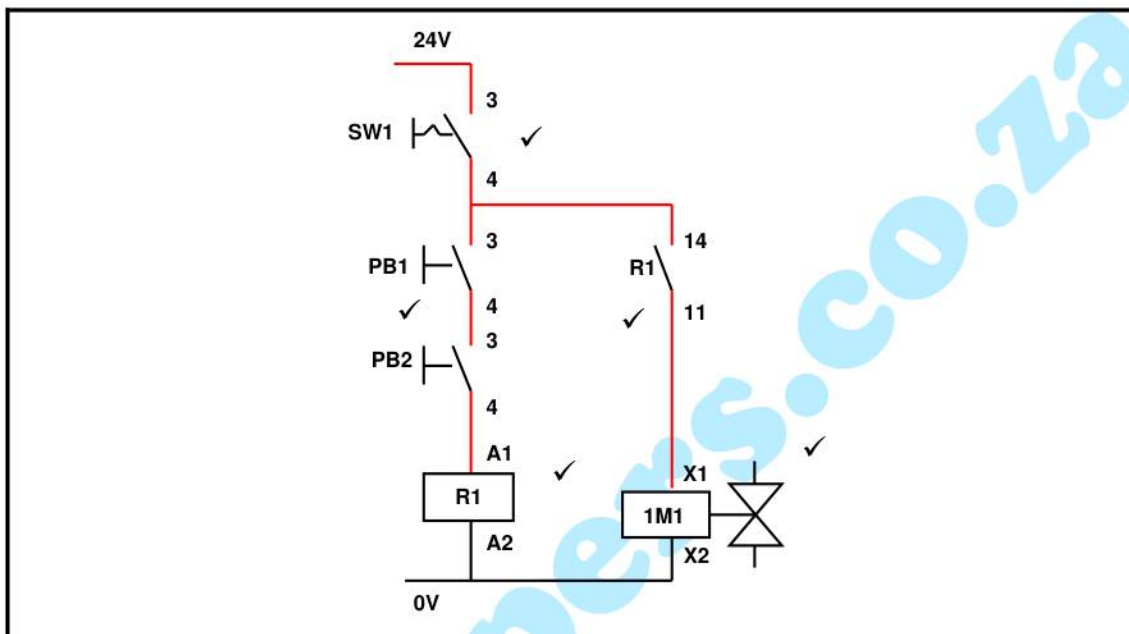
(6)
[20]

QUESTION 4

- 4.1
- 2 push buttons (NO)✓✓
 - Solenoid valve ✓
 - 24-V relay with changeover contacts ✓
 - Single-acting cylinder ✓

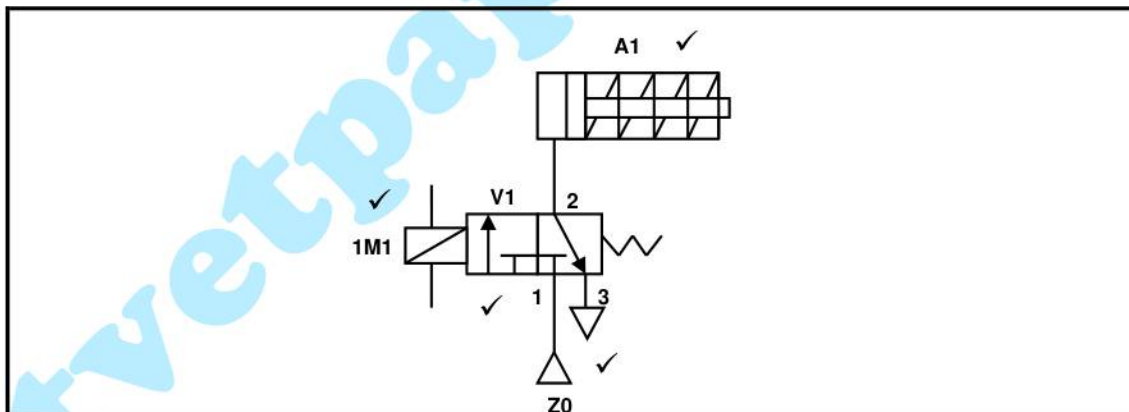
(5)

4.2



(5)

4.3



(4)

- 4.4
- The two-handed control is used to prevent any injuries such as crushing. The operation employs the indirect method of manual control,✓ where the control relay (R1) will control the solenoid valve (1M1). When PB1 and PB2 are depressed simultaneously, the cylinder will extend,✓ and when the two push buttons are released, the cylinder will retract.✓

(3)

- 4.5
- By adding a pre-set time-based timer
 - Adding a normally open timer contact in the control circuit
 - Adding a relay latching (retaining) (sustaining) contact

(3)

[20]

QUESTION 5

- 5.1 Troubleshooting is the process of locating a fault condition that causes a system to malfunction. (2)
- 5.2 Output – Check the output to see if there is supply. If there is, replace the output device; if not, to the next stage.
Input – Check to see if there is supply. If not, check the mains supply; if there is, check through the circuit.
Check through the circuit – Check in the middle of the circuit that is being tested: If there is supply, then check the second half of the circuit by applying the half-cut approach. If there is no supply, apply the half-cut method to the first half of the circuit. (3 × 2) (6)
- 5.3
- Live circuit testing is done on a circuit that is energised using a multimeter set on volts.
 - Dead circuit testing is done using a continuity tester on a circuit that has been isolated (locked out) from the supply. (2 × 2) (4)
- 5.4
- 5.4.1 Faulty solenoid valve✓ – Replace the valve.✓
- 5.4.2 Check the supply at the input to the circuit.✓
Apply throughout the circuit testing.✓
- 5.4.3 Check the pneumatic circuit using the output-input-through-the-circuit approach.✓✓
- 5.4.4 Check the cylinder for damage.✓
Check the timer control circuit for operation.✓
(4 × 2) (8)
- [20]**
- TOTAL: 100**